Vraj Prajapati

Aerospace Engineering Student | CAD and Design for Manufacturing | U.S. Citizen
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EXPERIENCE

Mechanical Engineering Intern, Raytheon Vision Systems — Goleta, CA | Jan 2026 – Present

- Designed and prototyped tooling for semiconductor wafer fabrication that moved the process from single-wafer to multi-wafer runs, increasing yield and productivity.
- Modeled in Creo, prototyped concepts on an SLA printer, and applied DFMA to turn designs into production-ready parts across multiple tooling revisions.
- Developed and built a new seal-off/rough-out station that removed a bottleneck and improved safety and precision in semiconductor growth operations.

Engineering Intern, Dos Pueblos Engineering Academy — Goleta, CA | Aug 2024 – Dec 2025

- Co-led a team of five, as technical contributor and machinist, in rebuilding 21 hobby mills: remade many parts to improve safety, reliability and ease of use for student operators.
- Machined precision parts on Hardinge lathes, manual mills and a Tormach CNC; modeled components in SolidWorks and wrote part drawings and technical write-ups.
- Worked the mechanical assembly station on an assembly line we built for the 21 mills, and helped route wiring harnesses.
- Built 3D-printed physics projects for STEM outreach to elementary school students.

Summer Engineering Intern, iCRco Inc. — Goleta, CA | Jun 2023 – Aug 2023

- Worked with assembly teams on CAT scan machines, upholding quality standards in regulated medical device production.
- Managed inventory workflows and supported shipping and receiving, improving accuracy and reducing discrepancies.

PROJECTS

Senior Capstone: Origami Penguin Folder, Dos Pueblos Engineering Academy — Aug 2023 – Jun 2024

- Led the half-fold mechanism for a four-person team: CAD, machining, assembly and testing of a custom miter gearbox and roller system. Three connected prototypes completed the first three folds.
- Traced a reversed-rotation failure found in testing (rollers ejected the paper) to shaft orientation in my own model, then redesigned and remade the shaft.

EDUCATION

California State Polytechnic University, Pomona, B.S. Aerospace Engineering — Pomona, CA | Aug 2026 – Jun 2028

- GPA 3.67. THRUST propulsion sub-team (MET and Resistojet research). Member, Society of Asian Scientists and Engineers.

Santa Barbara City College, Six Associate Degrees with Honors — Santa Barbara, CA | Aug 2022 – Jun 2026

- GPA 3.5. Engineering, Mathematics, Mathematics (AS-T), Physics (AS-T), and two in Liberal Arts and Sciences.
- Payload and Recovery Lead, Aerospace and Rocket Club: completed Level 1 and Level 2 launches and designed an automated parafoil recovery payload (Fusion 360, MATLAB).

SKILLS

CAD and Design: SolidWorks (assemblies, drawings), Creo, Fusion 360, MATLAB, OpenRocket
Manufacturing: CNC milling and turning (Haas, Tormach), manual machining (Hardinge lathe, mill), 3D printing (FDM, SLA), laser cutting, DFMA
Electronics and Mechatronics: Soldering, PCB design, Arduino